Econ 240B: Lecture 2

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January 27, 2025

This lecture covers sample average as an OLS interpretation.

1 Sample Averages (in Gaussian Location Model)

Recall that the sample average is given by

$$\bar{Y} = \frac{1}{N}\sum_{i=1}^{N} Y_i$$

where $Y_i \sim$ iid $\mathcal{N}(\mu, \sigma)$ with σ^2 known and μ unknown. There are several ways to interpret this sample average.

1.1 Oracle ML

Suppose we assumed we knew information about the distribution we actually didn't. In this case, we are assuming that σ^2 is known. Then

$$\bar{Y} = \tilde{\mu}_{ML}$$

where

$$\begin{aligned}\tilde{\mu}_{ML} &= \arg\max_{\mu_Y}\left\{-\frac{N}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{N}\left(Y_i - \mu_Y\right)^2\right\} \\ &= \arg\min_{\mu_Y}\sum_{i=1}^{N}\left(Y_i - \mu_Y\right)^2\end{aligned}$$

Thus, solving for the Oracle ML estimator by maximizing the log-likelihood function is equivalent minimizing the sum of squares above. These types of estimators are also called least squares estimators.

1.2 OLS Interpretation of $\bar{Y}$

The least squares estimator for μ is given by

$$\bar{Y} = \hat{\mu}_{OLS}$$

where

$$\hat{\mu}_{OLS} = \arg\min_{\mu_Y}\sum_{i=1}^{N}\left(Y_i - \mu_Y\right)^2$$

We can reframe the problem and consider the following location model,

$$Y_i = \mu + \varepsilon_i$$

where $\varepsilon \sim \text{iid } \mathcal{N}\left(0, \sigma^2\right)$ defines the unexplained part of μ is the explained part of Y_i which is constant.

The interpretation here is that we seek to minimize the unexplained part. Functionally, the estimator, $\hat{\mu}_{OLS}$, minimizes the implied length of $\varepsilon = (\varepsilon_1, ..., \varepsilon_N)$

$$||\varepsilon|| = \sqrt{\sum_{i=1}^{N} \varepsilon_i^2}$$

2 Properties of $\bar{Y}$ Under Gaussian Location Model

For the following section, assume that we are working with the following location model

$$Y_i = \mu + \varepsilon_i, \varepsilon_i \sim \text{iid } \mathcal{N}\left(0, \sigma^2\right)$$

2.1 Statistical/Probability Properties of $\bar{Y}$

The mean of our sample average is

$$\mathbb{E}_\mu\left[\bar{Y}\right] = ... = \mu, \ \ (\forall \mu)$$

therefore, $\bar{Y}$ is unbiased. The variance is

$$\text{Var}_\mu\left(\bar{Y}\right) = ... = \frac{\sigma^2}{N}, \ \ (\forall \mu)$$

therefore, the Mean-Squared Error (MSE) is

$$\begin{aligned} MSE\left(\bar{Y}\right) &= \mathbb{E}\left[\left(Y_i - \bar{Y}\right)^2\right] \\ &= \text{Var}_\mu\left(\bar{Y}\right) + \text{Bias}_\mu\left(\bar{Y}\right)^2 \\ &= \text{Var}_\mu\left(\bar{Y}\right) + \left(\mathbb{E}_\mu\left[\bar{Y}\right] - \mu\right)^2 \\ &= \frac{\sigma^2}{N}, \ \ (\forall \mu) \end{aligned}$$

2.2 Statistical Properties of $\bar{Y}$

- Sufficiency
- Completeness
- Efficiency: uniform minimum variance unbiased estimator (UMVU)

2.3 Distributional Properties of $\bar{Y}$

Give our assumptions on Y_i, we know that

$$\bar{Y} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{N}\right)$$

We can redefine the above statement as either:

1. **Centering of $\bar{Y}$/ Variance Stabilization**
   (a) $\sqrt{N}\left(\bar{Y} - \mu\right) \sim \mathcal{N}\left(0, \sigma^2\right)$
2. **Standardization**
   (a) $\frac{\bar{N}(\bar{Y}-\mu)}{\sigma} \sim \mathcal{N}\left(0, 1\right)$

2.4 Inference of μ

Oftentimes, we want to understand how "good" of an estimate we're working with. We can do this by consider the confidence interval and the associated coverage probability that the "true" mean falls somewhere close to our estimate. For any $c \geq 0$, let

$$\mathrm{CI}(\mu) = \left[\bar{Y} - c\frac{\sigma}{\sqrt{N}}, \bar{Y} + c\frac{\sigma}{\sqrt{N}}\right]$$

where $\frac{\sigma}{\sqrt{N}}$ is the standard deviation of $\bar{Y}$. Then $\mathrm{CI}(\mu)$ is a confidence interval for μ with a coverage probability,

$$\begin{aligned}
P_\mu(\{\mu \in \mathrm{CI}(\mu)\}) &= P_\mu\left(\bar{Y} - c\frac{\sigma}{\sqrt{N}} \leq \mu \leq \bar{Y} + c\frac{\sigma}{\sqrt{N}}\right) \\
&= P_\mu\left(-c \leq \frac{\sqrt{N}\left(\bar{Y} - \mu\right)}{\sigma} \leq c\right) \\
&= \Phi(c) - \Phi(-c) \\
&= 2\Phi(c) - 1
\end{aligned}$$

where Φ is the CDF of a standard normal. Therefore, if $c = -\Phi^{-1}(0.025)$, we get the 95% confidence interval which tells us that if we were to repeat the sampling process and construct the CI each time, then 95% of these intervals will have the true mean.

3 Properties of $\bar{Y}$ Under General iid Location Model

Now consider the location model that is iid, but where the errors are not necessarily normally distributed.

$$Y_i = \mu + \varepsilon_i,\ \varepsilon_i \sim \text{iid } \left(0, \sigma^2\right)$$

then

$$Y_i \sim \text{iid } \left(\mu, \sigma^2\right)$$

3.1 Interpretation of $\bar{Y}$ (as an estimator of μ)

Recall all of the previous interpretation of our sample average in the previous lecture.

- ML: There are ways to maximize log-likelihood of joint densities as a function of unknowns. We will still get that $\mu_{ML} = \bar{Y}$.
- "Oracle" ML: VML but assuming we have knowns. Can reverse engineer to to maximize the log-likelihood of distribution w.r.t to parameters and the implied Oracle ML is also just $\bar{Y}$.
- "Quasi" ML: Variant of ML with heroic assumptions are likely to be wrong to see what happens when you maximize likelihood w.r.t to incorrect parameters. OLS is an example of Quasi ML with $\bar{Y} = \bar{\mu}_{ML}$ when the reference distribution is normal (w or w/o known variance – the assumption is made on the errors).
- GMM: Again, we have $\bar{Y} = \hat{\mu}_{GMM}$, where

  $$\frac{1}{N}\sum_{i=1}^{N} g\left(Y_i, \hat{\mu}_{GMM}\right) = 0$$

  with

  $$g\left(Y, \mu_Y\right) = Y - \mu_Y$$

Recall that if a parameter is correct then in expectation,

$$\mathbb{E}\left[g\left(Y_i, \mu_Y\right)\right] = \mathbb{E}\left[Y_i\right] - \mu_Y = \mu - \mu_Y \Leftrightarrow \mu_Y = \mu$$

The GMM requires less distributional assumptions than ML.

- "Plug-in"/ "Analogous": For any cdf $F_Y : \mathbb{R} \to [0,1]$, we have the operator when applied at the cdf,

$$\mu\left(F_Y\right) = \int_{-\infty}^{\infty} y dF_Y = \arg\min \int_{-\infty}^{kn} \left(y - \mu_Y\right)^2 dF_Y\left(y\right)$$

then

$$\mu\left(F\right) = ... = \mu$$

and

$$\mu\left(\hat{F}_N\right) = ... = \bar{Y}$$

where $\hat{F}_N$ is the empirical cdf.

- OLS:

$$\bar{Y} = \hat{\mu}_{OLS} = \arg\min_{\mu_Y} \sum_{i=1}^{N} \left(Y_i - \mu_Y\right)^2$$

3.2 Statistical/Probabilistic Properties of $\bar{Y}$

Even after relaxing the assumption of normality, we still have that

$$\mathbb{E}_\mu\left(\bar{Y}\right) = ... = \mu \ \ (\forall \mu)$$

therefore, μ is unbiased. We also have that

$$\mathrm{Var}_\mu\left(\bar{Y}\right) = ... = \frac{\sigma^2}{N} \ (\forall \mu)$$

so the results from earlier are robust to the distribution so long as we preserve iid.

3.3 Statistical Properties of $\bar{Y}$

The statistical properties outlined earlier are not robust to distribution

3.4 Distributional Properties of $\bar{Y}$

Theorem 1. *Cramer's Deconvolution Theorem.*

$$\bar{Y} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{N}\right) \Leftrightarrow Y_i \sim iid\, \mathcal{N}\left(\mu, \sigma^2\right)$$

Cramer's Deconvolution Theorem tells us that the desired properties from earlier for CI construction are not robust. We ca rephrase the theorem above as the following two equivalent statements,

Theorem 2. *Iff $Y_i \sim iid\, \mathcal{N}\left(\mu, \sigma^2\right)$, then*

$$P\left(\frac{\sqrt{N}\left(\bar{Y} - \mu\right)}{\sigma} \leq t\right) = \Phi\left(t\right), \ \forall t \in \mathbb{R} \tag{1}$$

$$\sup_{t \in \mathbb{R}} \left| P\left(\frac{\sqrt{N}\left(\bar{Y} - \mu\right)}{\sigma} \leq t\right) - \Phi\left(t\right) \right| = 0 \tag{2}$$

However, even if the observations are not normally distribution, it turns that that $\bar{Y}$ is approximately $\mathcal{N}\left(\mu, \frac{\sigma^2}{N}\right)$ distributed for "large" N "in general".

Here is the following fact:

Fact 3. *If $Y_i \sim$ iid $\left(\mu, \sigma^2\right)$, then*

$$\lim_{n\to\infty} P\left(\frac{\sqrt{N}\left(\bar{Y}-\mu\right)}{\sigma} \leq t\right) = \Phi\left(t\right), \ \forall t \in \mathbb{R} \tag{3}$$

$$\lim_{n\to\infty} \sup_{t\in\mathbb{R}} \left| P\left(\frac{\sqrt{N}\left(\bar{Y}-\mu\right)}{\sigma} \leq t\right) - \Phi\left(t\right)\right| = 0 \tag{4}$$

where (4) *is uniform convergence in probability and implies point-wise convergence in probability* (3). (4) $\Rightarrow$ (3) *always, but it turns out that we will also have* (3) $\Rightarrow$ (4).

3.5 Implication

What kind of conclusions can we draw if we take (3) and (4) at face value? It turns out that we can actually say something about the coverage probabilities as we were able to do in the Gauss Location model. Recall that the confidence interval is given by

$$\mathrm{CI}\left(\mu\right) = \left[\bar{Y} - c\frac{\sigma}{\sqrt{N}}, \bar{Y} + c\frac{\sigma}{\sqrt{N}}\right]$$

with coverage probability

$$P\left(\left\{\mu \in \mathrm{CI}\left(\mu\right)\right\}\right) = \Phi\left(c\right) - \Phi\left(-c\right)$$

For large N of $Y_i \sim$ iid $\left(\mu, \sigma^2\right)$. Then

$$P\left(\left\{\mu \in \mathrm{CI}\left(\mu\right)\right\}\right) \to_N \Phi\left(c\right) - \Phi\left(-c\right) = 1 - 2\Phi\left(-c\right)$$

and again pick c s.t. $P\left(\left\{\mu \in \mathrm{CI}\left(\mu\right)\right\}\right) = 0.95$.

This is a nice property especially since the asymptotics for the type of data we care about start kicking in at around $N = 30$ to $N = 50$.